EA ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2021 Worked Solutions

January 2021 paper rebuilt with the worked solution after each question.

10
paper questions

WMA12
Pure 2

Single-paper solutions
standalone IAL route

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P2

PAST PAPER

WMA12/01 January 2021

January 2021 | 10 questions | 75 marks

10
questions

75
marks

Answers
worked solution after each question

Question 1

6 marks

Polynomials

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1. $$f(x) = x^4 + ax^3 - 3x^2 + bx + 5$$

where a and b are constants.

When $f(x)$ is divided by $(x + 1)$, the remainder is 4

(a) Show that $a + b = -1$

(2)

When $f(x)$ is divided by $(x - 2)$, the remainder is -23

(b) Find the value of a and the value of b.

(4)

(Total 6 marks)

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Worked Solution - Question 1

Polynomials

Topic group

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: -- therefore -- = - therefore = 5, 4 a b = - =

Final answer

-- therefore -- = - therefore = 5, 4 a b = - =

Question 2

7 marks

Applications of Differentiation

blank

2. A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

(a) Using calculus, find the x coordinates of the stationary points of the curve.

(4)

(b) Justify, by further calculus, the nature of all of the stationary points of the curve.

(3)

(Total 7 marks)

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Worked Solution - Question 2

Applications of Differentiation

Topic group

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: a minimum

Final answer

a minimum

Question 3

8 marks

Laws of Logarithms

blank

3. (i) Solve

$$7^{x+2} = 3$$

giving your answer in the form $x = \log_7 a$ where a is a rational number in its simplest form.

(3)

(ii) Using the laws of logarithms, solve

$$1 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$$

(5)

(Total 8 marks)

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Worked Solution - Question 3

Laws of Logarithms

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: therefore = d 1 2 y =

Final answer

therefore = d 1 2 y =

Question 4

8 marks

Binomial Expansion

blank

4. (a) Find the first three terms, in ascending powers of x, of the binomial expansion of

$$(2 + px)^6$$

where p is a constant. Give each term in simplest form.

(4)

Given that in the expansion of

$$\left(3 - \frac{1}{2}x\right)(2 + px)^6$$

the coefficient of x^2 is $-\frac{3}{4}$

(b) find the possible values of p.

(4)

(Total 8 marks)

Worked Solution - Question 4

Binomial Expansion

Topic group

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: therefore = d ( ) 1 1 , 8 120 p =

Final answer

therefore = d ( ) 1 1 , 8 120 p =

Question 5

4 marks

Proof

5. (i) Use algebra to prove that for all $x \geqslant 0$

$$3x + 1 \geqslant 2\sqrt{3x}$$

(3)

(ii) Show that the following statement is not true.

"The sum of three consecutive prime numbers is always a multiple of 5"

(1)

(Total 4 marks)

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Worked Solution - Question 5

Proof

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: not true for three consecutive prime numbers Eg 5 7 11 23 + + = which is not divisible by 5 (so not true)

Final answer

not true for three consecutive prime numbers Eg 5 7 11 23 + + = which is not divisible by 5 (so not true)

Question 6

8 marks

Trigonometric Equations

blank

6. (a) Show that the equation

$$\frac{3\sin\theta\cos\theta}{2\sin\theta - 1} = 5\tan\theta \qquad \sin\theta \neq \frac{1}{2}$$

can be written in the form

$$3\sin^3\theta + 10\sin^2\theta - 8\sin\theta = 0$$

(4)

(b) Hence solve, for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\frac{3\sin 2x\cos 2x}{2\sin 2x - 1} = 5\tan 2x$$

giving your answers to 3 decimal places where appropriate.

(4)

(Total 8 marks)

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Worked Solution - Question 6

Trigonometric Equations

Topic group

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: 0 x =

Final answer

0 x =

Question 7

6 marks

Integration

blank

7.

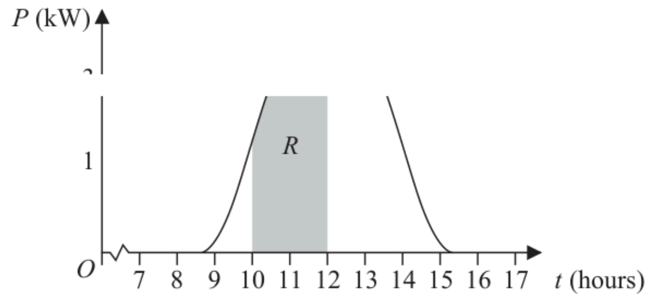

Figure 1

Solar panels are installed on the roof of a building.

The power, P, produced on a particular day, in kW, can be modelled by the equation

$$P = 0.95 + 2^{t-12} + 2^{12-t} - (t-12)^2 \qquad 8.5 \leqslant t \leqslant 15.2$$

where t is the time in hours after midnight. The graph of P against t is shown in Figure 1.

A table of values of t and P is shown below, with the values of P given to 4 significant figures where appropriate.

Time, t (hours)	10	10.5	11	11.5	12
Power, P (kW)		1.882	2.45		2.95

(a) Use the given equation to complete the table, giving the values of P to 4 significant figures where appropriate.

(2)

The amount of energy, in kWh, produced between 10:00 and 12:00 can be found by calculating the area of region R, shown shaded in Figure 1.

(b) Use the trapezium rule, with all the values of P in the completed table, to find an estimate for the amount of energy produced between 10:00 and 12:00. Give your answer to 2 decimal places.

(4)

(Total 6 marks)

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Worked Solution - Question 7

Integration

Topic group

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: [ ] 1 1 " " "1

Final answer

[ ] 1 1 " " "1

Question 8

7 marks

Sequences & Series

blank

8. A sequence $a_1, a_2, a_3, \ldots$ is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$
$$a_1 = p - 3$$

where p is a constant.

(a) Find an expression for a_2 in terms of p, giving your answer in simplest form.

(1)

Given that $\sum_{n=1}^{3} a_n = p + 15$

(b) find the possible values of a_2

(6)

(Total 7 marks)

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Worked Solution - Question 8

Sequences & Series

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: = therefore + Possible values for 2a are 1 7 and 2 -

Final answer

= therefore + Possible values for 2a are 1 7 and 2 -

Question 9

10 marks

Circles

blank

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

(a) Write down, in terms of k,

(i) the coordinates of the centre of C,

(ii) the radius of C.

(2)

Given that the point $P(2, 3)$ lies on C

(b) (i) show that $5k^2 - 17k + 6 = 0$

(ii) hence find the possible values of k.

(3)

The tangent to the circle at P intersects the x-axis at point T.

Given that $k < 2$

(c) calculate the exact area of triangle OPT.

(5)

(Total 10 marks)

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Worked Solution - Question 9

Circles

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: × × = × + × 147 16 = oe

Final answer

× × = × + × 147 16 = oe

Question 10

11 marks

Sequences & Series

blank

10. In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan A, he will increase the distance he runs each Sunday by the same amount.

(a) Calculate the distance he will run on Sunday in week 5 under training plan A.

(3)

In training plan B, he will increase the distance he runs each Sunday by the same percentage.

(b) Calculate the distance he will run on Sunday in week 5 under training plan B.
Give your answer in km to one decimal place.

(3)

Owen will also run a fixed distance, x km, each Wednesday over the 12-week period.

Given that

- x is an integer
- the total distance that Owen will run on Sundays and Wednesdays over the 12 weeks will not exceed 360 km

(c) (i) find the maximum value of x, if he uses training plan A,

(ii) find the maximum value of x, if he uses training plan B.

(5)

Q10

(Total 11 marks)

(Total 11 marks)

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Worked Solution - Question 10

Sequences & Series

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: or \leq Training model A: x = 4 km, Training model B: x = 5km cso

Final answer

or \leq Training model A: x = 4 km, Training model B: x = 5km cso